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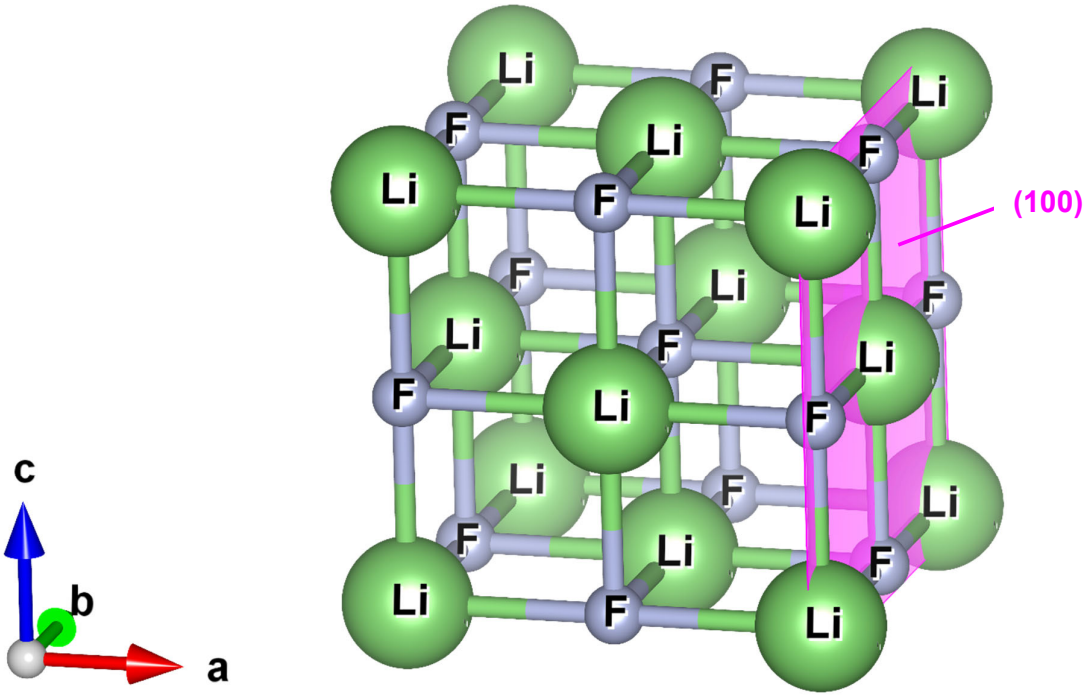
College of Science

Department of Physics

Fourth Year (Undergraduate Students)

Solid State Physics Lab.

Manual



First Semester

2026-2027

Names of the Experiments

Experiment 1: Temperature Coefficient of Resistance in Semiconductors

Experiment 2: X-Ray Diffraction (XRD) Crystal Structure Investigation of LiF

Experiment 3: X-Ray Diffraction (XRD) – Moseley's Law

Experiment 4: X-Ray Powder Diffraction (XRPD) – Crystal Structure Investigation of GaAs

Experiment 5: Electron Diffraction – Crystal Structure Investigation of Graphite

Experiment 6: Semiconductor Thermoelectric Generator

Experiment 1

Temperature Coefficient of Resistance in Semiconductors

Purpose

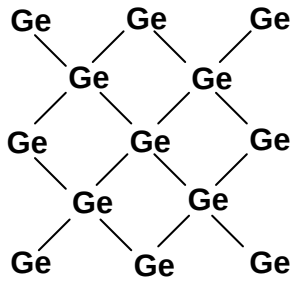
1) To show temperature dependence of resistance of Germanium.

2) Determine its temperature coefficient of resistance.

3) To show temperature dependence of the temperature coefficient of resistance.

Theory

Semiconductors are materials that fall midway between conductors and insulators. Their valence electrons are neither as free as those in a conductor nor as tightly bound as those in an insulator.



Diamond like crystal structure of pure germanium

Fig. 1

The valence electrons are actually shared by the atoms in the semiconductor resulting in covalent bonding Fig. 1. Covalent bonding permits electrons to move from atom to atom without being entirely free. Semiconductor materials, such as silicon and germanium, have only small numbers of conduction electrons.

The resistance of a semiconductor decreases as the temperature increases, as shown in Fig. 2. The effect is opposite to and much greater than the change in the resistance of a conductor (metal) with temperature.

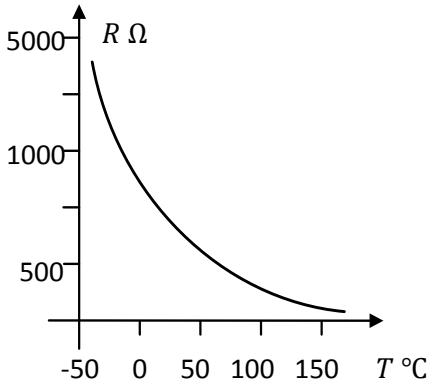


Fig. 2

As the temperature increases, the thermal energy of the valence electrons may become greater than their binding energy to the nuclei. The covalent bond is then broken, the electron releases and becomes a free electron, as a result a hole is produced and thereby increasing the number of carriers.

On the other hand, as in metals, a temperature rise increases the amplitude of vibration of atoms in the crystal structure (the lattice), which therefore collide more frequently with moving charge carriers and hence increase in resistance. But the effect of this which tends to increase resistance is overwhelmed in semiconductors by the effect of the release of more and more charge carriers (electrons and holes) as the temperature increases.

Over all, the resistance R of a pure semiconductor, theoretically, varies with absolute temperature T according to the exponential relationship:

$$R = R_o e^{\left(\frac{E_g}{2k_B T}\right)} \qquad (1)$$

Where R_o (the resistance at 0 K) and E_g (the energy gap) are constants for a given semiconductor and k_B is the Boltzmann constant.

Also the resistance of the semiconductor can be calculated from the electrical current (*I*) passing through it and the potential difference (*V*) across it:

$$R = \frac{V}{I} \qquad (2)$$

A quantity to measure change of electrical resistance *dR* of any substance (conductors, semiconductors, and insulators) with the change of temperature *dT* is called temperature coefficient of resistance, denoted $\boldsymbol{\alpha}$. It has dimension of inverse temperature, and is defined as follows:

$$\frac{dR}{R} = \alpha \, dT \qquad (3)$$

Value of temperature coefficient of resistance depends on three factors: type of material, temperature, and the initial temperature on which the increment of resistance is based. However, If the temperature coefficient itself does not vary too much with temperature and ($\alpha \Delta T \ll 1$), a linear approximation will be useful in estimating the value *R* at a temperature *T* (aka $R_{(T)}$), given its value R_0 at a reference temperature T_0. Integrating Eq. 3:

$$\int_{R_o}^{R_{(T)}} \frac{dR}{R} = \int_{T_o}^{T} \alpha \, dT$$

$$\Rightarrow \quad R_{(T)} = R_o e^{\alpha (T - T_o)} \qquad (4)$$

Applying the Taylor series approximation at the first order, in the proximity of T_o leads to:

$$R_{(T)} = R_o(1 + \alpha(T - T_o))$$

$$\Rightarrow \alpha = \frac{R_{(T)} - R_o}{R_o(T - T_o)} \qquad (5)$$

For strongly temperature-dependent α, this approximation is only useful for small temperature differences ΔT.

Procedure

1. Connect the apparatus as shown in Fig. 4.
2. Set the current to a value of $5mA$.
3. Set the display in the temperature mode.
4. Start the measurement by activating the heating coil with the on/off-knob on the back side of the board.



Fig. 4

5. Let the temperature rise to about 140℃, then (if it did not turn off automatically) turn the heater off and allow the system to cool down.
6. Record the temperature and the corresponding voltage on the voltmeter down to about 40℃ every 10℃.
7. Tabulate your data as shown below:

$\boldsymbol{T}_{(\boldsymbol{C}^{\circ})}$	$\boldsymbol{V}_{(\boldsymbol{volt})}$	$\boldsymbol{R}_{(\boldsymbol{\Omega})} = \boldsymbol{V}/\boldsymbol{I}$
140		
130		
. . . 40	. . .	

Calculation

1) Plot a graph of the values of R as ordinates against corresponding values of T as abscissa (Fig. 5), then take 6 different temperatures on the curve and their corresponding values of resistance. From these find 6 values of temperature find coefficient of resistance $\boldsymbol{\alpha}$ using Eq. 5.

Tabulate your calculations as follows:

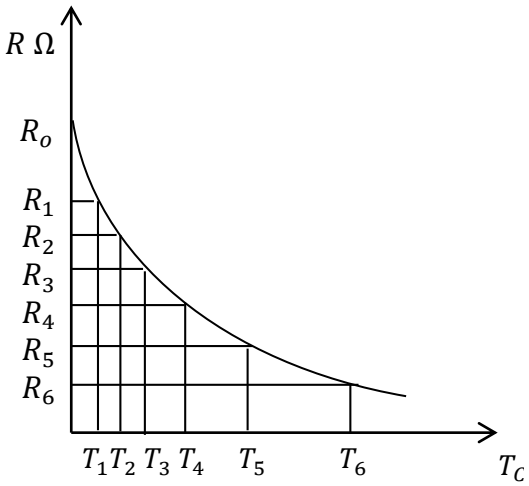


Fig. 5

$\boldsymbol{T}$ (℃)	$\boldsymbol{\alpha}$ (℃)$^{-1}$
T_1	$\alpha_1 = \frac{R_1 - R_o}{R_o(T_1 - 0)}$
T_2	$\alpha_2 = \frac{R_2 - R_o}{R_o(T_2 - 0)}$
T_3	$\alpha_3 = \frac{R_3 - R_o}{R_o(T_3 - 0)}$
.	.
.	.
.	.

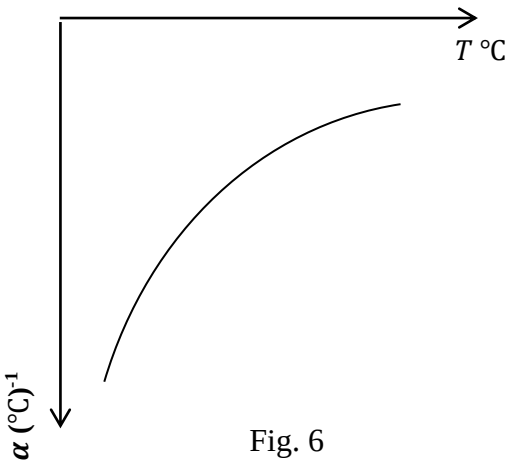


Fig. 6

2) Plot a graph for this new table between the values of the temperature T as abscissa and its corresponding values of temperature coefficient of resistance α as ordinates (Fig. 6).

Questions

1. Explain how increasing the temperature of a metal increases its resistance.
2. Explain how increasing the temperature of a semiconductor decreases the resistance.

Experiment (2)

X-Ray Diffraction (XRD) - Crystal Structure Investigation of LiF

Purpose

To determine the lattice constant and interplanar spacing of (100) LiF monocrystal by analyzing the wavelength-dependent intensity of the diffracted X-ray.

Theory

When electrons of high energy impinge on the metallic anode of an X-ray tube, X-rays with a continuous energy distribution (the so-called bremsstrahlung) are produced. X-ray lines whose energies are not dependent on the anode voltage, and which are specific to the anode materials, the so-called characteristic X-ray lines, are superimposed on the continuum. They are produced as follows:

An impact of an electron on an anode atom in the K shell, for example, can ionize that atom. The resulting vacancy in the shell is then filled by an electron from a higher energy level. The energy released in this de-excitation process can then be transformed into an X-ray which is specific for the anode atom. Characteristic X-rays produced from either the L → K or the M → K transitions are called Kα and Kβ lines respectively. M1 → K and L1 → K transitions do not take place due to quantum mechanical selection rules.

Accordingly, characteristic lines for Cu with the following energies are to be expected:

$$E_{K\alpha^*} = E_K - 1/2(E_{L2} + E_{L3}) = 8.038\ keV \qquad (1)$$

$$E_{K\beta} = E_K - E_{M2.3} = 8.905\ keV \qquad (2)$$

K_{α^*} is used as the mean value of the lines $K_{\alpha 1}$and $K_{\alpha 2}$.

Bragg's law of reflection describes the diffraction of plane waves at a monocrystal as the selective reflection of the waves at a set of lattice planes within the crystal. Due to the periodicity of the crystal, the lattice planes of a set have a fixed spacing d(hkl). An incident wave with the wavelength λ is reflected with maximum intensity when the Bragg condition

$$n\lambda = 2d \sin\theta \qquad (3)$$

where n is the diffraction order and d is the spacing of lattice planes.

The angle θ shows the direction of the incident and reflected wave respect to the set of lattice planes and is often referred to as the glancing angle.

In a cubic crystal with LiF (100) structure, the lattice planes run parallel to the crystal's unit cells in the simplest case. Their spacing d(hkl) corresponds to one lattice constant:

$$d_{100} = a_0 \qquad (4)$$

This lets us use Eq. (3) as an equation for determining the lattice constant a_o:

$$n\lambda = 2\, a_0 \sin\theta \qquad (5)$$

In other words, to determine a_o we need to measure the glancing angle θ for a known wavelength λ and diffraction order n. This method is more precise when the glancing angles are also measured in higher diffraction orders.

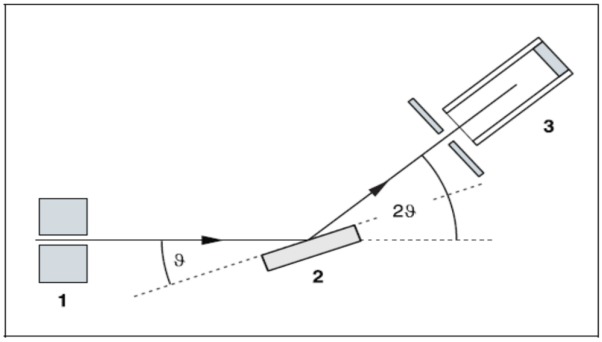


Fig. 1: Schematic diagram of diffraction of X-ray at a monocrystal and 2θ coupling between counter-tube angle and scattering angle (glancing angle) (1collimator, 2 monocrystal, 3 counter tube).

A Geiger-Muller counter tube (Type B) is used to detect the X-rays; this instrument and the crystal are both pivoted with respect to the incident X-ray beam in 2θ coupling the counter tube is truned by twice the angle of the crystal (Fig. 1).



Fig. 2: Experimental set-up for the examination of the structure of LiF monocrystals.

Procedure

1) Switch on the X-ray unit
2) Start the "Measure" program and select "Gauge" -> "X-ray Device"
3) Select the parameters shown in Fig. 3(a) and press the continue button (select the Crystal you are using: LiF).
4) Start the measurement (see Fig. 3(b) and Fig. 4)

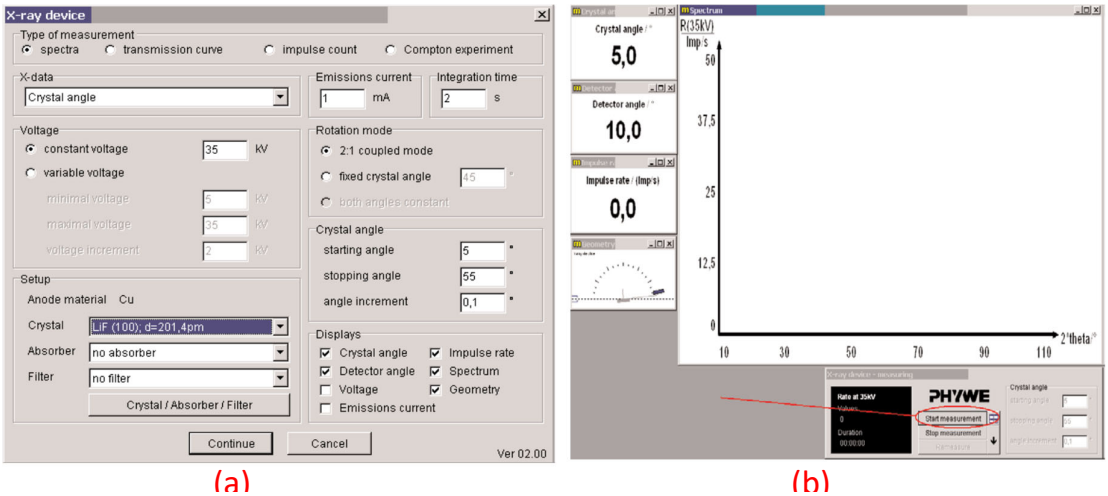


Fig. 3: (a) Measuring parameters for the recording software. (b) The graphical user interface of the "X-ray Device"-software.

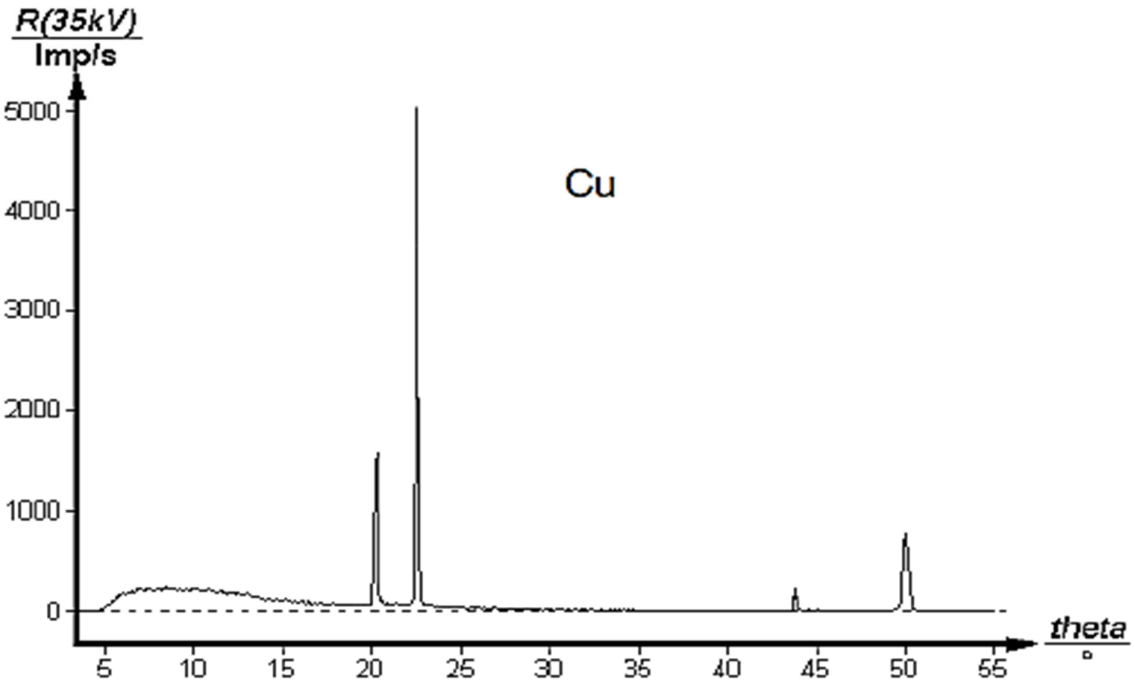


Fig. 4: X-ray spectrum of Copper (Z = 29). Analyzer crystal: LiF.

Calculations:

1) In your resulted spectrum locate the lines k_β and k_α, and evaluate the angle of each peak (the glancing angle).

2) For each glancing angle θ, calculate the values sinθ, and n · λ as shown in the following table:

Table 1: X-ray diffraction data of LiF crystal

Line	n	λ (pm)	$n\lambda$ (pm)	θ (degree)	$Sin\theta$
$\boldsymbol{k_\beta}$	1	139.23			
$\boldsymbol{k_\alpha}$	1	153.45			
$\boldsymbol{k_\beta}$	2	139.23			
$\boldsymbol{k_\alpha}$	2	153.45			

3) Draw a graph between the values of $\boldsymbol{n\lambda}$ as ordinates versus the corresponding values of $\boldsymbol{Sin\theta}$ as abscissa as shown in Fig. 4.

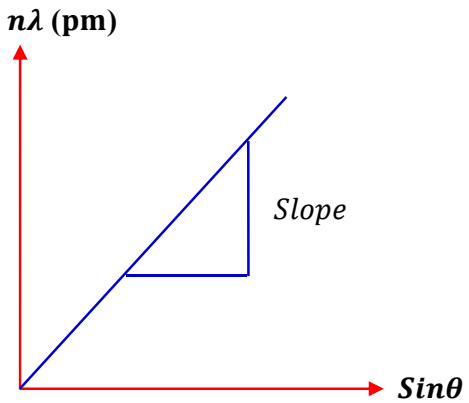


Fig. 4

5) From this graph, Eqs. (4) and (5), find the lattice constant a_o and the interplanar spacing of (100) planes:

$$a_o = \frac{slope}{2}$$

Questions

(1) Write Bragg's law and explain its significance.
(2) What is an X-ray?
(3) Why cannot ordinary optical grating diffract X-ray?
(4) Can normal light be used for crystal diffraction? If it is so, under what conditions?
(5) In X-ray diffraction, the angle θ is measured from the reflecting surface rather than from the normal? Why?
(6) Compare between Electron diffraction and X-ray diffraction.

Experiment 3

Moseley's Law

Purpose

1) Verifying Moseley's law.

2) Determining the Rydberg constant.

Theory:

In 1913 British physicist Moseley (Gwyn Jeffreys Moseley) investigated systematically the characteristic radiation emitted by different elements as X-ray targets. He determined the frequencies of the X-rays emitted from the different elements. Moseley then investigated the possibility of some relation between the frequency of a given spectral line (f) and some property of the atom emitting the line like atomic number. He plotted the square root of the frequencies of a given line against the atomic number, and not atomic weight. He found that this is a fundamental property of elements. This can be represented by the equation.

$$\sqrt{f} = a(Z - b) \qquad (1)$$

where Z is the atomic number of the absorbing element, a and b are constants depending upon the particular line. For the K series (k_α line) $b = 1$; and for K series:

$$a = \sqrt{\frac{3cR_\infty}{4}}$$

where R_∞ is the Rydberg constant. $c = 3 \times 10^8$ m/sec is the speed of light in free space. Thus Eq. 1 becomes;

$$\sqrt{f} = \sqrt{\frac{3cR_\infty}{4}}(Z - b) \qquad (2)$$

Also we have the Bragg equation,

$$n.\lambda = 2d \sin\theta \qquad (3)$$

where;

$n = 1, 2, 3, ...$ is the order number of an intensity maximum.

λ is the incident X-ray wavelength.

d is the crystal planes (or atomic) separation, or crystal inter planes (or interatomic) distance.

θ is the angle of diffraction (the angle at which scattered intensity is maximum).

The frequency of the incident X-ray relates to its wavelength through speed of light with the below relation;

$$f = c/\lambda \qquad (4)$$



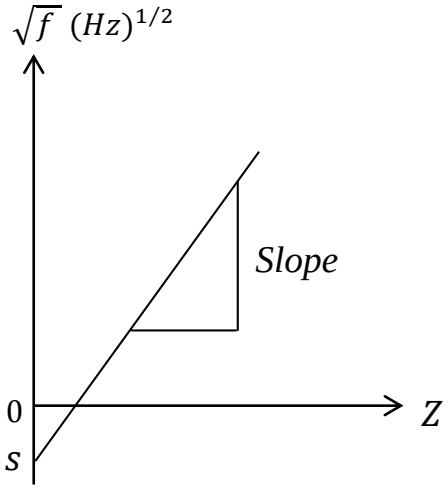
Experimental setup

Procedure

1- Arrange the apparatus as shown in Fig. 1.
2- With Fe X-ray tube is connected. Find the characteristic spectrum for Fe.
3- Repeat the above step for Cu and Mo or any other given X-ray tubes.
4- From the K_a characteristic spectrum find the Bragg angle (θ) for the (111) lattice plane, which corresponds to the first peak.
5- Find the wavelength of the k_α line using Eq. (2). Lattice constant of NaCl is 5.62 Angstroms.
6- Find the frequency of the k_α line using Eq (3).
7- Tabulate the readings as below:

Elements	Z	2θ (degree)	$\lambda = 2d_{111}\, sin\theta$	$f = c/\lambda$ (Hz)	$\sqrt{f}$ $(\text{Hz})^{1/2}$
K	19				
Ca	20				
:	:				
Zn	30				

8- Draw a graph for the values of square root of frequency of k_α line ($\sqrt{f}$) for the given elements against their corresponding atomic number (Z) as shown in Fig. 2.



Calculation

1- From the slope of Fig. 2 and Eq.2, find the Rydberg constant (R_∞).

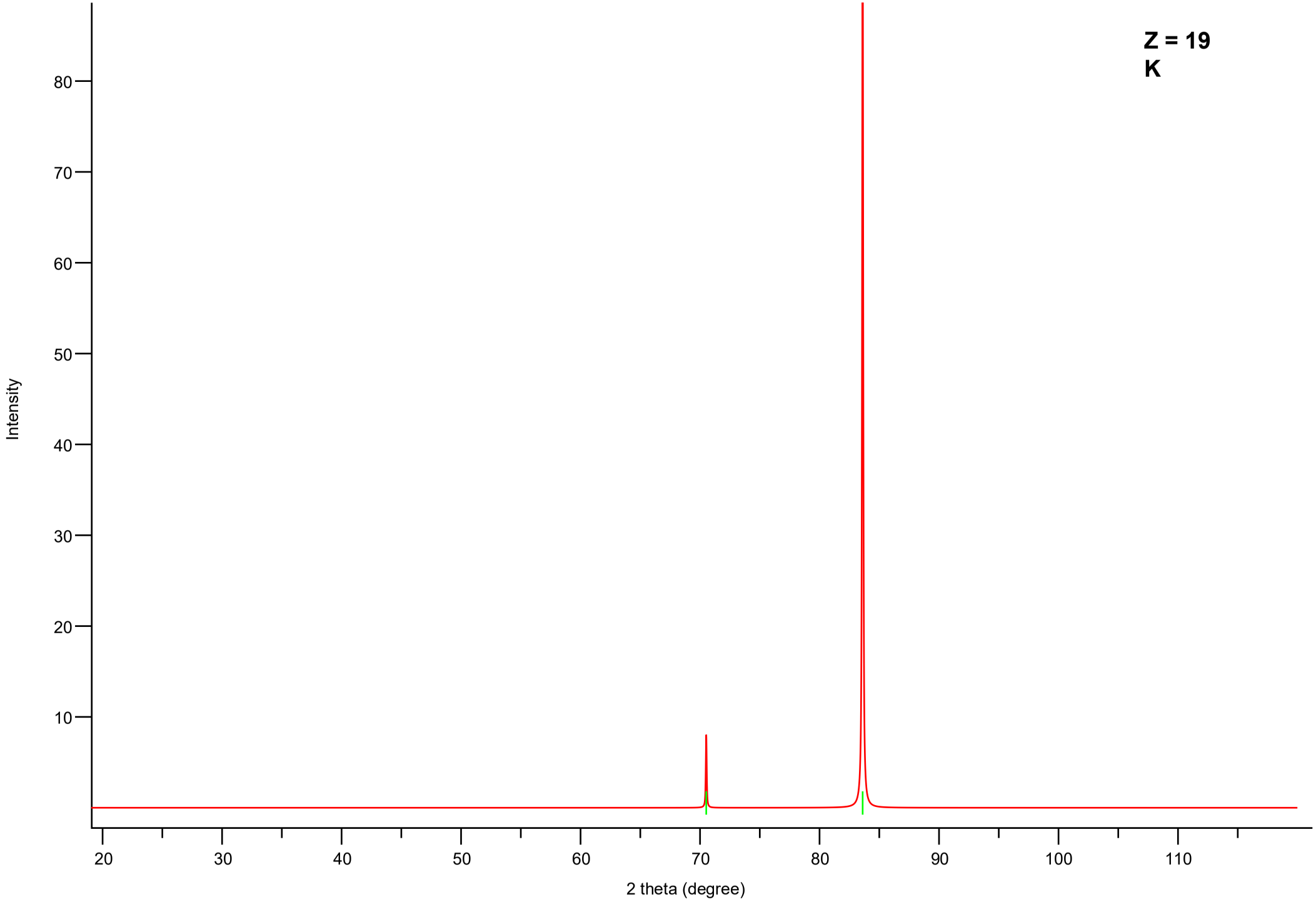
$$R_\infty = \frac{4}{3}\frac{(slope)^2}{c}$$

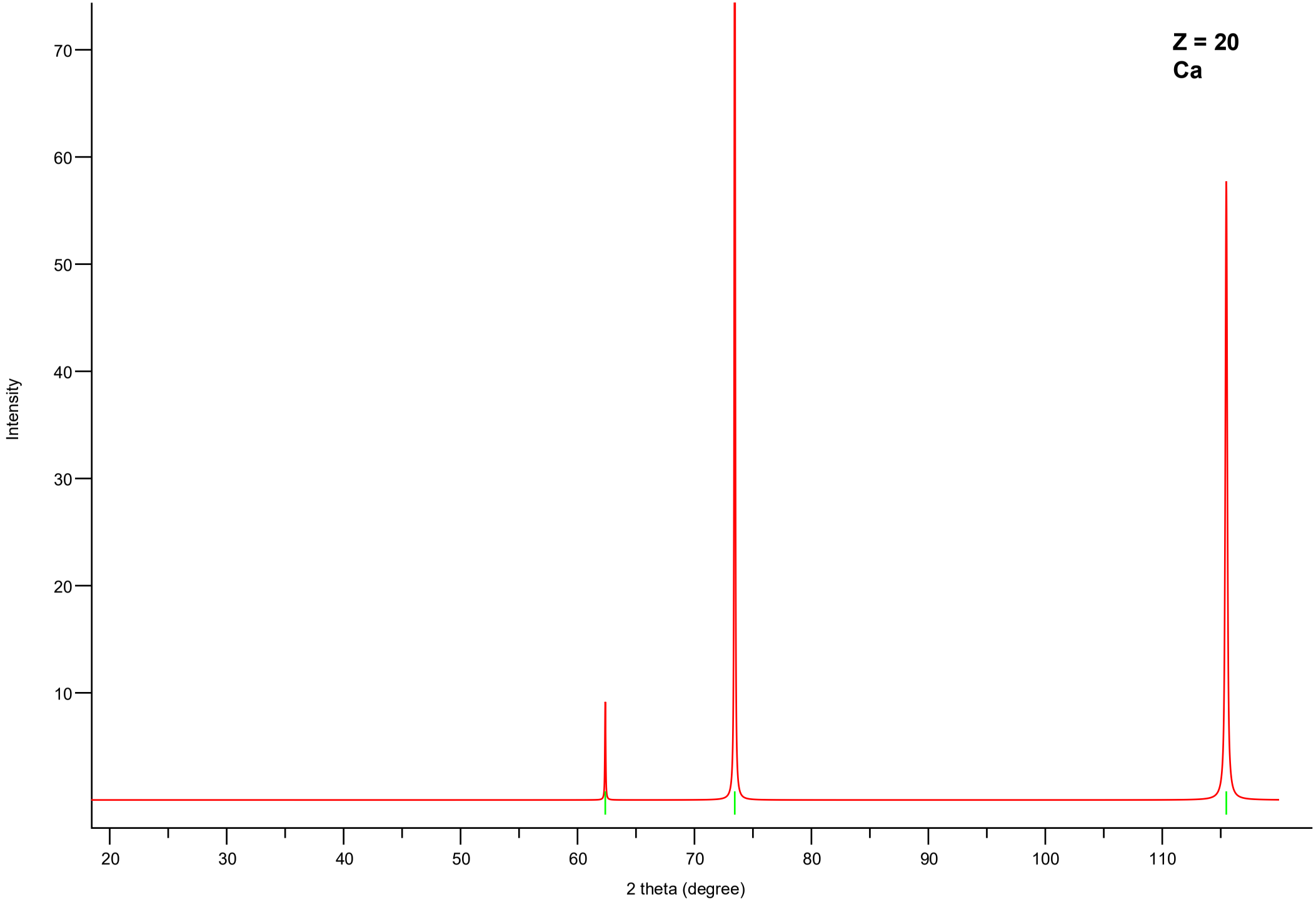
2- Find the b constant from the intersect:

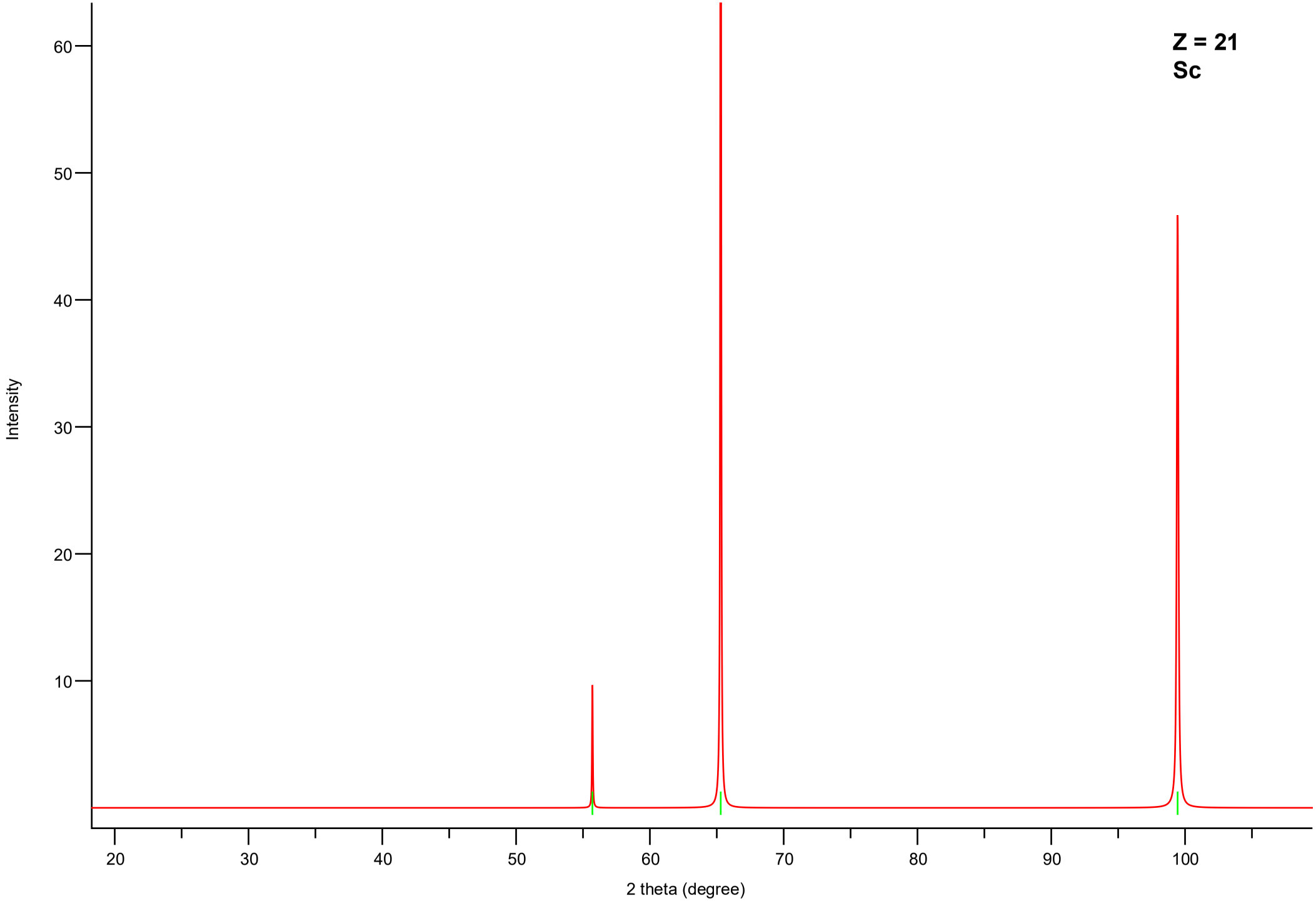
$$ab = 0s \Rightarrow b = 0s/\text{slope}$$

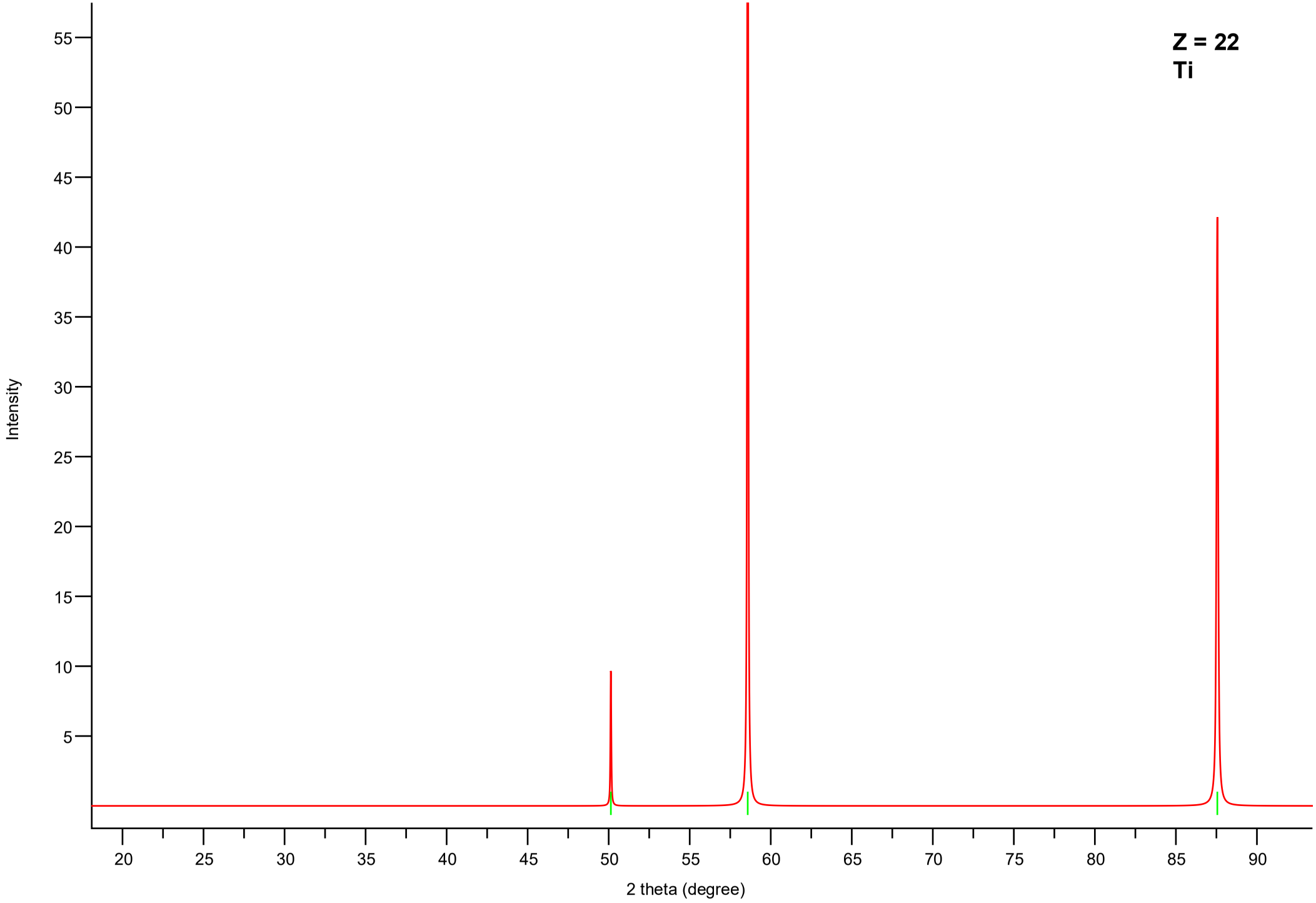
Questions:

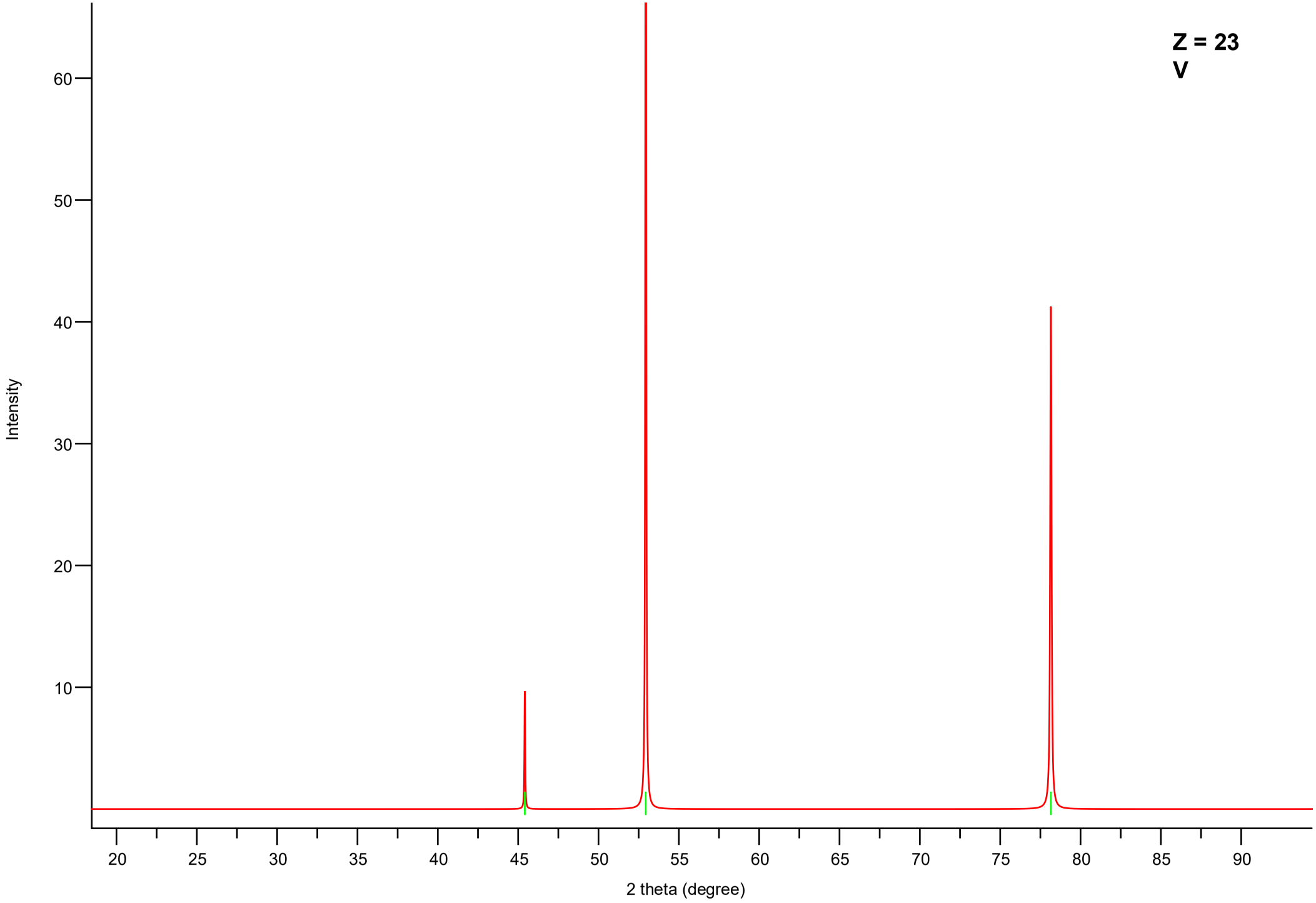
1- Distinguish between characteristic X- ray and bremsstrahlung X-ray.
2- What is the importance of Moseley's law?
3- Describe a method for the production of X-rays.
4- Derive Moseley's law from Bohr Theory of the atom.
5- Why the frequency of lines produced from an element of higher atomic number is greater than that produced by an element of low atomic number?

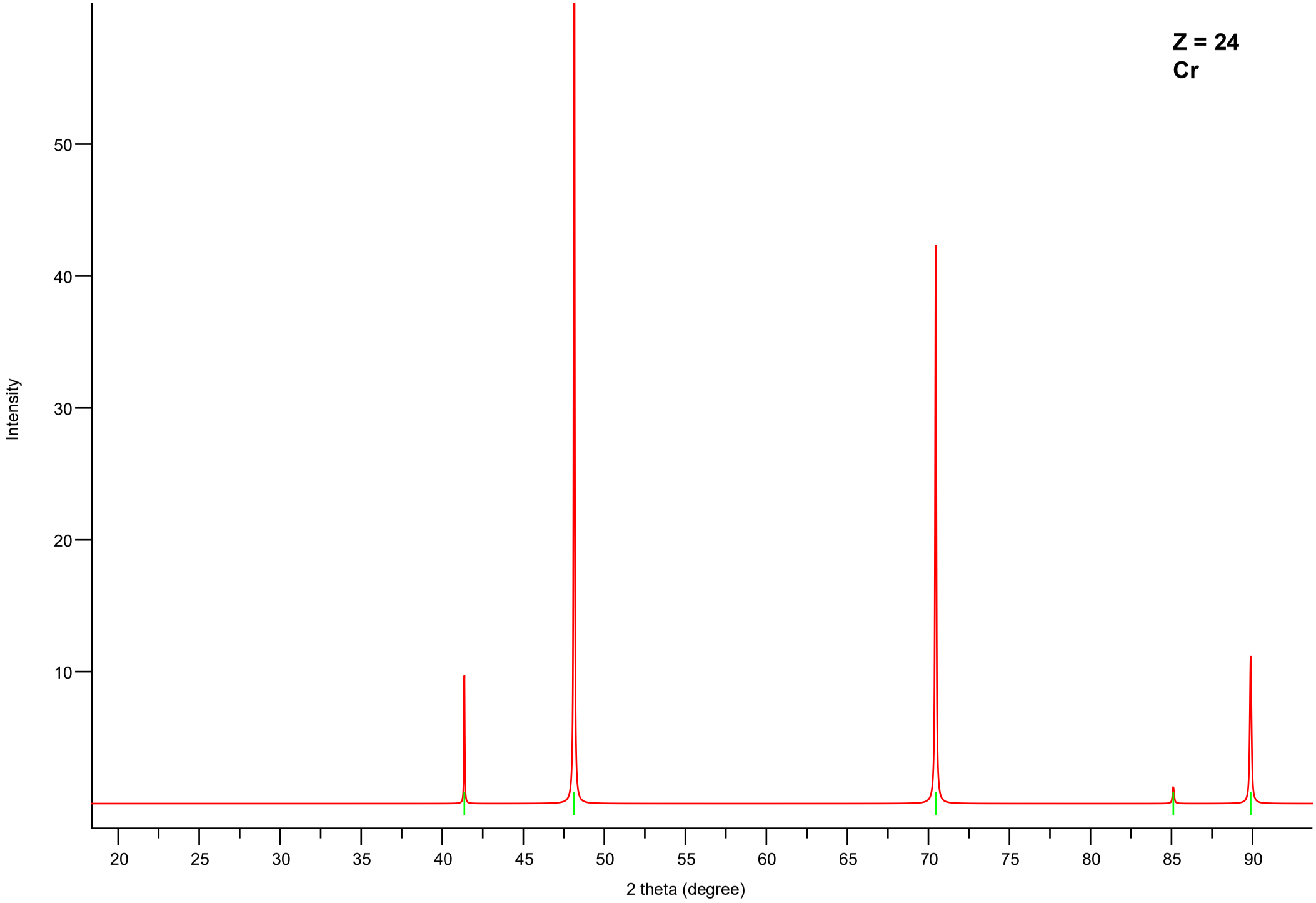

Z = 19
K
Intensity
80
70
60
50
40
30
20
10
20
30
40
50
60
70
80
90
100
110
2 theta (degree)

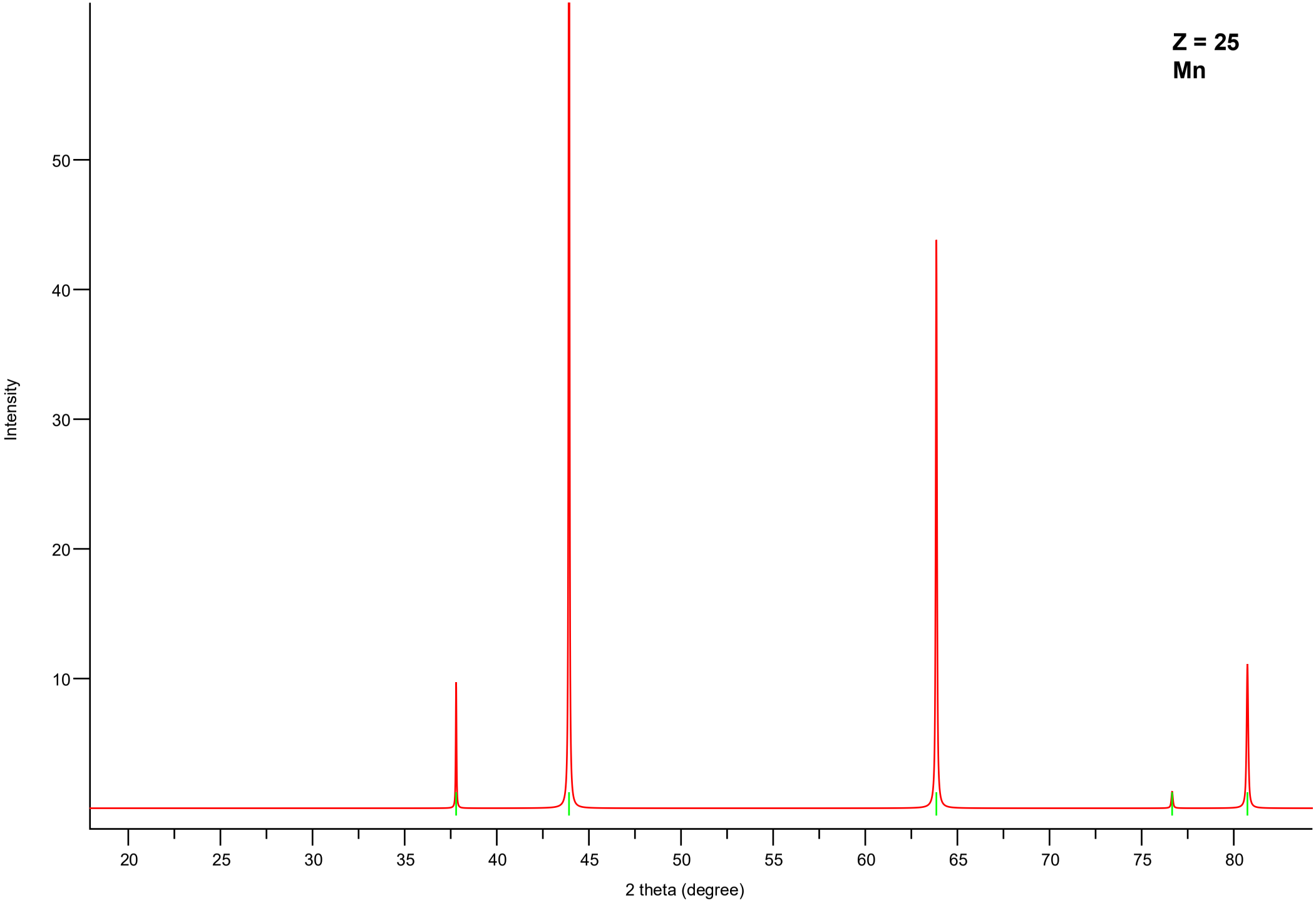

Z = 20
Ca
Intensity
70
60
50
40
30
20
10
20
30
40
50
60
70
80
90
100
110
2 theta (degree)

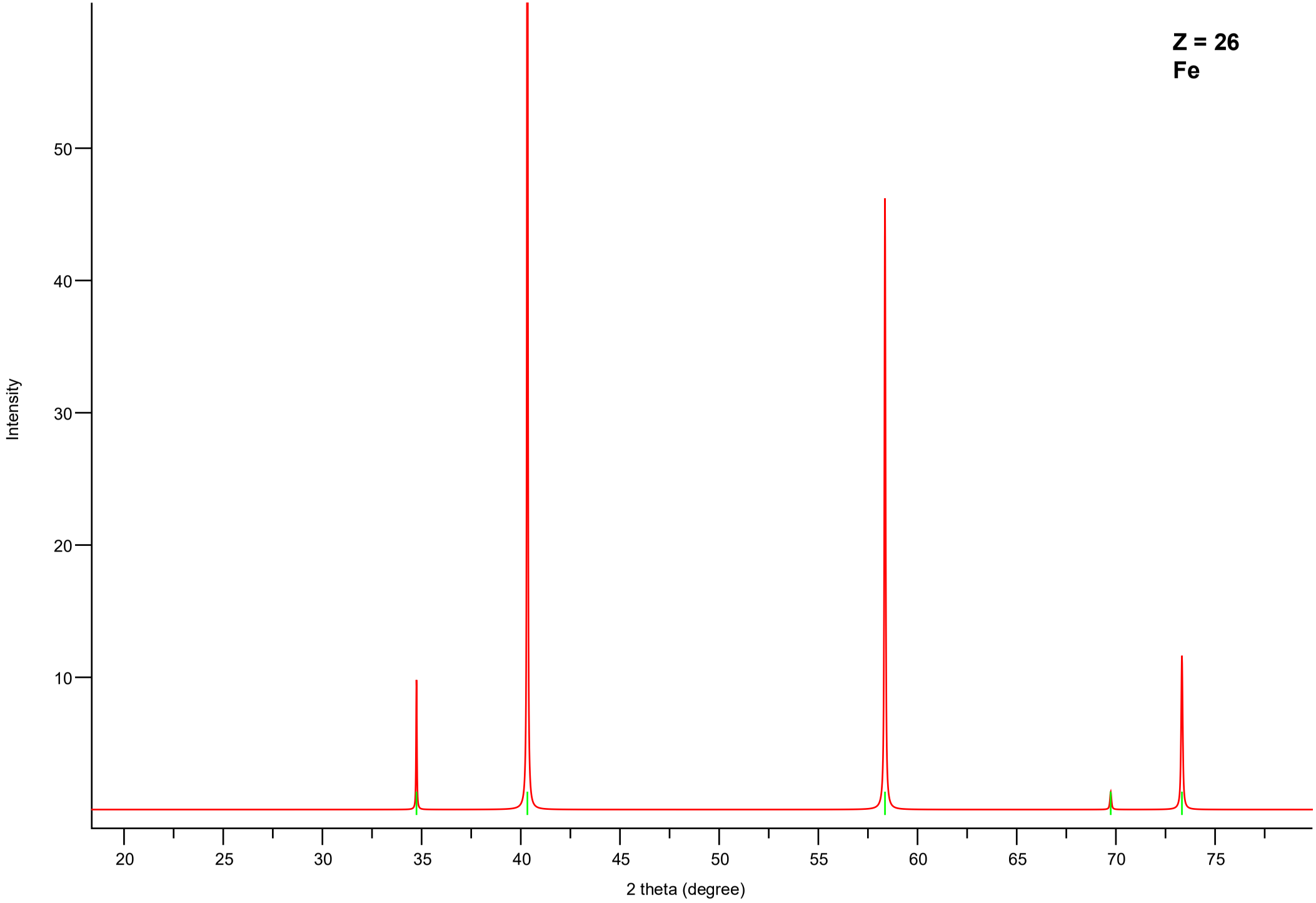

Z = 21
Sc
Intensity
60
50
40
30
20
10
20
30
40
50
60
70
80
90
100
2 theta (degree)

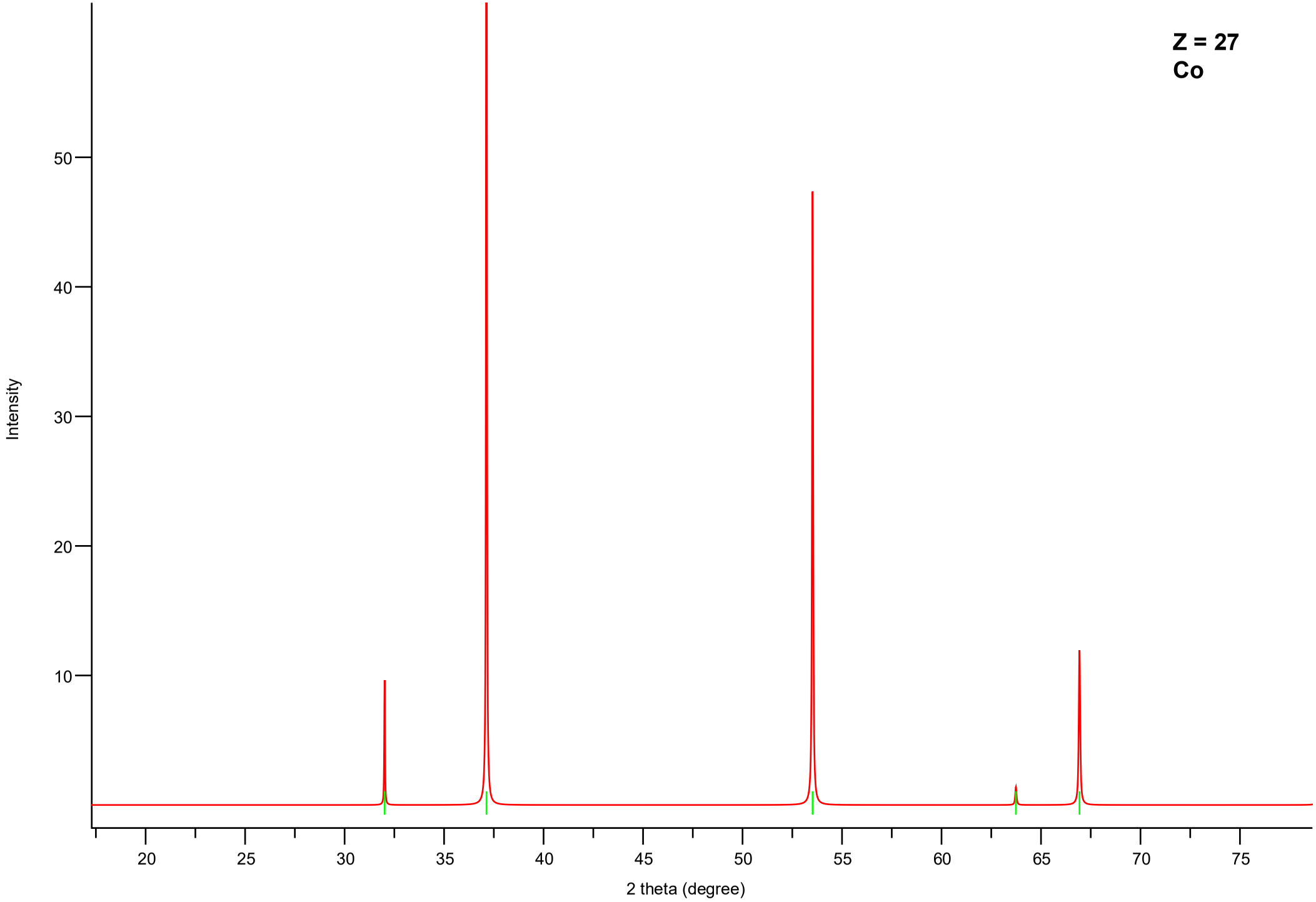

Z = 22
Ti
Intensity
55
50
45
40
35
30
25
20
15
10
5
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
2 theta (degree)

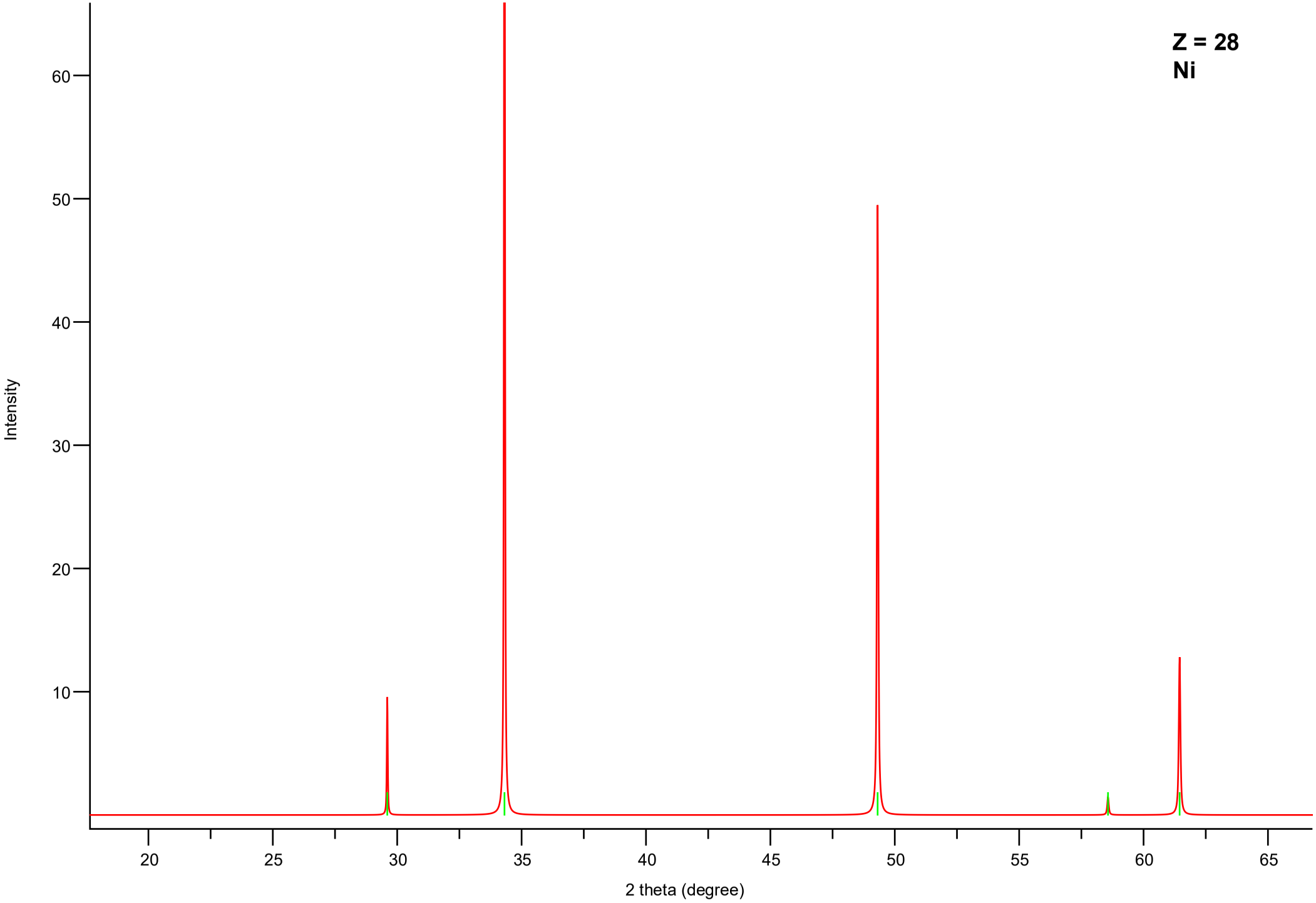

Z = 23
V
Intensity
60
50
40
30
20
10
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
2 theta (degree)

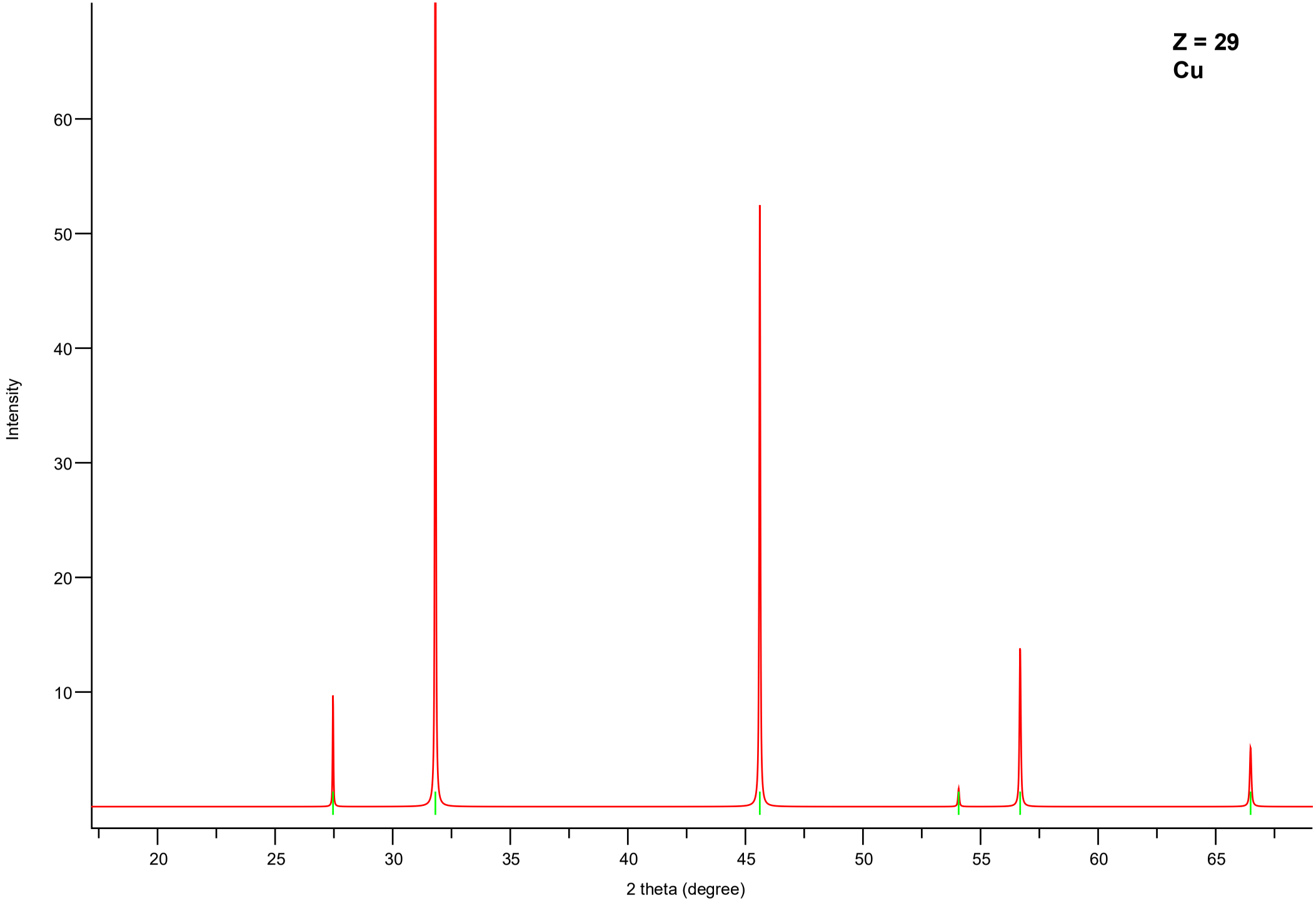

Z = 24
Cr
Intensity
50
40
30
20
10
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
2 theta (degree)

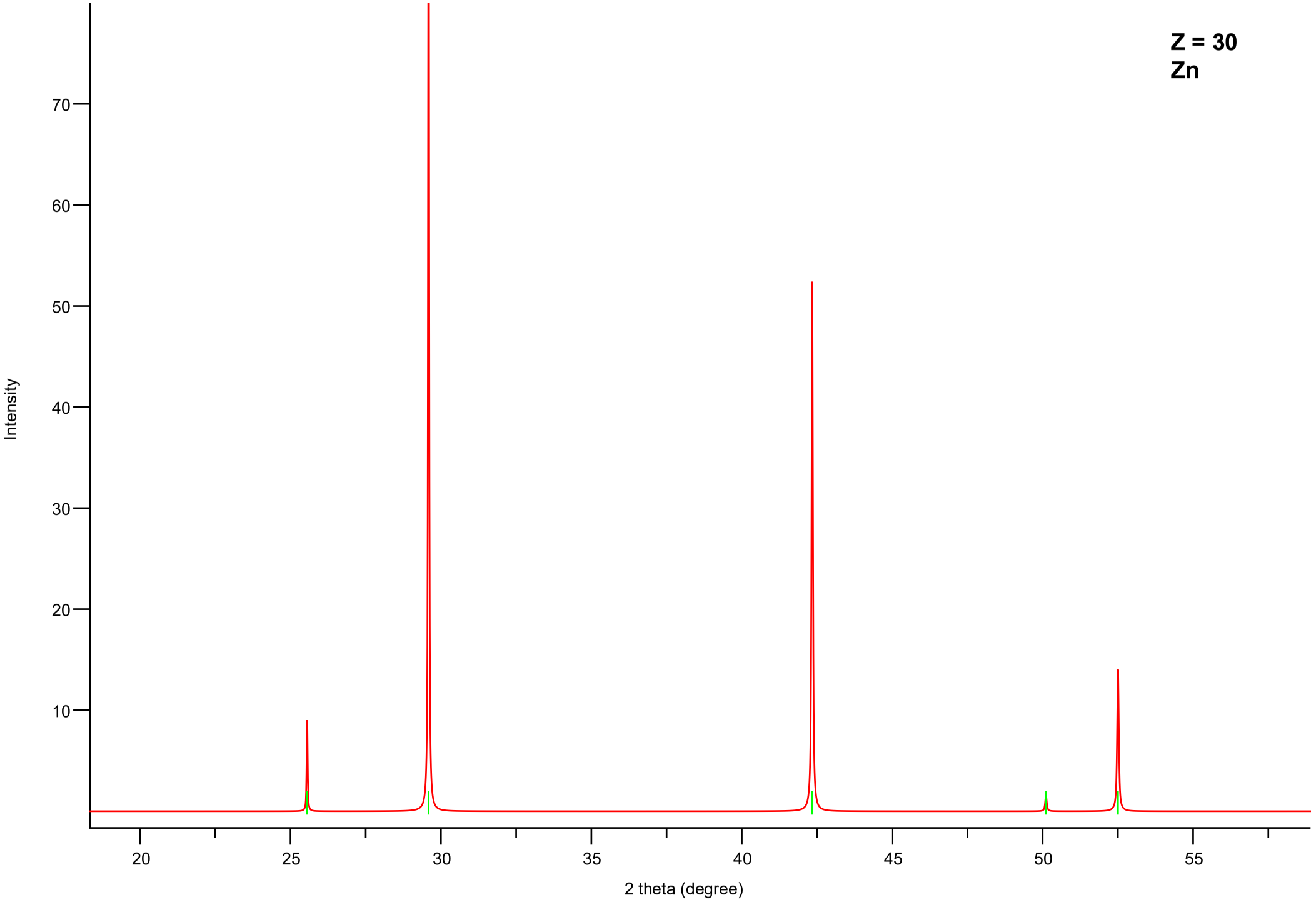

Z = 25
Mn
Intensity
50
40
30
20
10
20
25
30
35
40
45
50
55
60
65
70
75
80
2 theta (degree)


Z = 26
Fe
Intensity
10
20
30
40
50
20
25
30
35
40
45
50
55
60
65
70
75
2 theta (degree)


Z = 27
Co
Intensity
10
20
30
40
50
20
25
30
35
40
45
50
55
60
65
70
75
2 theta (degree)


Z = 28
Ni
Intensity
60
50
40
30
20
10
20
25
30
35
40
45
50
55
60
65
2 theta (degree)


Z = 29
Cu
Intensity
60
50
40
30
20
10
20
25
30
35
40
45
50
55
60
65
2 theta (degree)


Z = 30
Zn
Intensity
70
60
50
40
30
20
10
20
25
30
35
40
45
50
55
2 theta (degree)

Experiment (4)

X-Ray Powder Diffraction (XRPD) – Crystal Structure Investigation of GaAs

Purpose

1- Polycrystalline samples are to be irradiated by an X-ray beam and the resulting diffraction patterns are recorded on a film and evaluated.

2- The Debye-Scherer rings are to be evaluated and assigned to the corresponding lattice planes.

3. The lattice constant of the sample material is to be determined.

Theory:

The structure of all crystals is described in terms of a single periodic lattice but with a group of atoms attached to each lattice point or situated in each elementary parallelepiped. This group of atoms is called the basis; the basis is repeated in space to form the crystal.

We often use the primitive translation vectors to define the crystal axes $\vec{a}$, $\vec{b}$, $\vec{c}$ although nonprimitive crystal axes may be used when they are more convenient or simpler. The crystal axes $\vec{a}$, $\vec{b}$, $\vec{c}$ form three adjacent edges of a parallelepiped. If there are lattice points only at the corners of the parallelepiped, then it is a primitive parallelepiped. Figure 1 shows the crystal axes.

There are fourteen types of space lattices. They are conveniently grouped into seven systems according to the seven types of unit cells; triclinic, monoclinic, orthorhombic, tetragonal, cubic, trigonal, and hexagonal.

For a cubic crystal $a = b = c$ and $\alpha = \beta = \gamma = 90^{o}$. It is useful for structure analysis to specify the orientation of a crystal plane by Miller indices. They are determined by:

(i) Finding the intercepts on the axes $\vec{a}$, $\vec{b}$, $\vec{c}$ in terms of the lattice constants. The axes may be primitive or nonprimitive.

(ii) Taking the reciprocals of these numbers and then reducing them to three integers having the same ratio, usually the smallest three integers. The result is enclosed in parentheses: ($\boldsymbol{h\ k\ l}$). The indices ($\boldsymbol{h\ k\ l}$) may denote a single plane or a set of parallel planes. If a plane cuts an axis on the negative

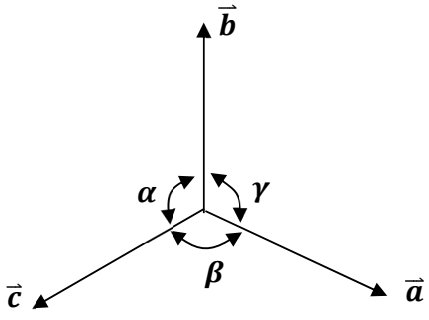


Fig. 1

side of the origin, the corresponding index is negative and is indicated by placing a minus sign above the index: (***h k l***).

Let us consider a beam of monochromatic X-rays incident on a set of parallel lattice planes spaced equal distances (d) apart (Fig. 2). The radiation is incident in the plane of the paper. The path difference for rays reflected from adjacent planes is ($2dsin\theta$), where (θ) which is Bragg angle (glancing angle) is measured from the plane. Constructive interference of the radiation reflected from successive planes occurs whenever the path difference is an integral number (n) of wavelengths (λ). Thus the condition for constructive reflection of the incident radiation is:

$$2d\ sin\theta = n\,\lambda \qquad (1)$$

This is Bragg's law (n is the order of diffraction). Observe that although the reflection from each plane is assumed to be specular (angle of incidence is equal to the angle of reflection), only for certain values of (θ) will the reflections from all parallel planes add up in phase to give a strong reflected (diffracted) beam.

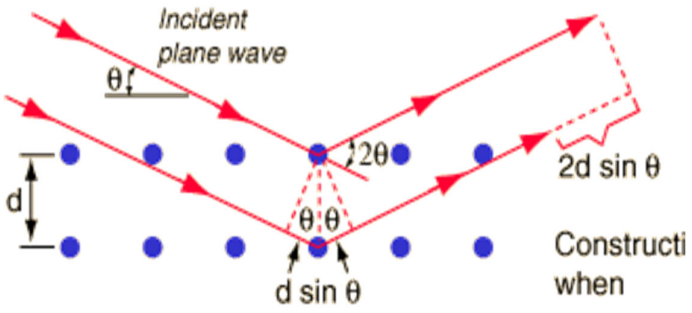


Fig. 2

The Bragg law is a consequence of the periodicity of the space lattice. The law does refer to the arrangement or basis of atoms associated with each lattice point.

The composition of the basis determines the relative intensity of the various orders (n) of diffraction from a given set of parallel planes.

Since $\sin\theta$ cannot exceed unity, the minimum spacing for which the reflection can be observed is $d = \lambda/2$ (i.e., $\lambda \leq 2d$). This is why we cannot use visible light!

For a cubic crystal with a lattice constant (a), the following relation is valid:

$$d(hkl) = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \qquad (2)$$

The Powder Method:

Since large, well-formed crystals are not always available, much X-ray diffraction work is done by making use of the powder method. This method was independently developed by Debye and Scherer (1916) and by Hule (1917).

In this method, the powdered sample containing a large number of small crystals is taken in a thin-walled glass capillary tube and irradiated with a beam of monochromatic X-rays. Since there are various minute crystals the orientation of the crystals will be random in all possible directions. But still, some crystals will be oriented in such a way that the Bragg equation holds for certain planes. Since there would always exist different orientations, for such crystals, parallel to a common axis, the X-rays are diffracted in such a way as to give a series of cones as shown in Fig. 3.

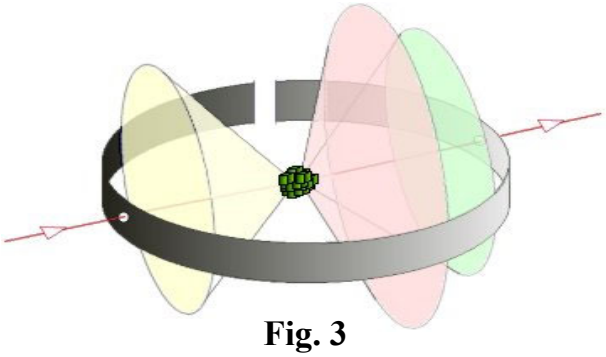


Fig. 3

The diffracted beam cones are allowed to fall onto a strip of circularly placed film. Each cone makes the image of two arcs (or lines) at the positions where the cone cuts the film. (See Fig. 4). The distance (R) between the specimen and the film (the radius) is determined.

It is well evident from what has been seen above that the lines in this diagram are really portions of an arc. Every symmetrical pair of lines in the film around the bright position at the center of the incident un-diffracted beam corresponds to a particular lattice plane of the crystal. From the distance L between these arcs and the center of the diffraction pattern and the radius (R), the angle of maximum reflection (θ) is determined. Thus, the spacing between lattice planes can be calculated.

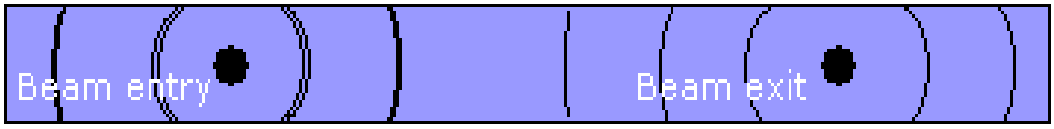


Fig. 4

The Bragg angle (glancing angle) for each arc (or line) related to the radius of the arcs (r) and the radius of the circular film (R) with the relation below:

$$\theta(\text{rad}) = \frac{1}{2}\frac{r}{R} \qquad (3)$$

Procedure

1- The X-ray diffraction for the sample under investigation is first obtained for a known X-ray wavelength on the film as shown in Fig. 4.

2- Numerate the arcs (or lines) from 1 to the end number from the nearest arc to the center and determine the index $\boldsymbol{hkl}$ for each arc.

3- Measure the distance ($\boldsymbol{r}$) between successive arcs and the center of the diffraction pattern, that is the radius of the arcs.

4- Measure the distance between the sample and film precisely, that is the radius of the circular film ($\boldsymbol{R}$), so that the exact diameter of the rings required for calculating the corresponding Bragg angles can be later determined.

5- Measure and record the Bragg angle (diffraction angle or glancing angle) ($\boldsymbol{\theta}$) using Eq. (3), and then find the $\boldsymbol{sin\theta}$ for each $\boldsymbol{\theta}$.

6- Determine the parallel lattice planes apart ($\boldsymbol{d}$) for the first order (i.e. $\boldsymbol{n = 1}$) using Eq. (1).

7- Tabulate your data (for **GaAs**) as follows:

Arc No.	$\boldsymbol{hkl}$	$\boldsymbol{\frac{1}{\sqrt{h^2+k^2+l^2}}}$	$\boldsymbol{r}$	$\boldsymbol{\theta_{rad} = \frac{1}{2}\frac{r}{R}}$	$\boldsymbol{\theta_{deg} = \frac{180}{\pi}\theta_{rad}}$	$\boldsymbol{\sin\theta}$	$\boldsymbol{d = \frac{\lambda}{2\sin\theta}}$
1	(111)						
2	(220)						
3	(311)						
4	(400)						
5	(331)						

$\boldsymbol{R} =$ cm

$\boldsymbol{\lambda = 1.6}$ Å

Calculations:

1- Draw a graph between the values of $\boldsymbol{d}$ as ordinates against the corresponding values of $\mathbf{1}/\sqrt{\boldsymbol{h^2 + k^2 + l^2}}$ as abscissa as shown in Fig. 5.

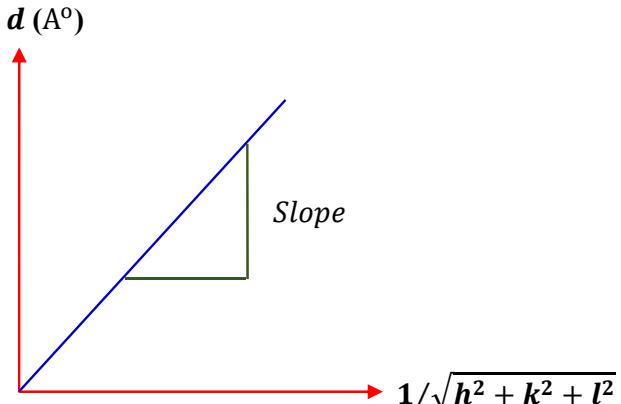


Fig. 5

2) From the slope of the graph of Fig. 5 and Eq. (2) find the lattice constant of the cubic crystal sample:

$$\boldsymbol{a = slope}$$

Questions:

1- The powder method of X-ray diffraction is widely used. Why?

2- For what purpose this technique is used?

3- Why monochromatic X-ray is used in the powder method?

Experiment 5

Electron Diffraction

Purpose

To determine the interplanar spacing of graphite.

Theory

To explain in the interference phenomenon, a wavelength λ, which depends on momentum, is assigned to the electrons in accordance with the de Broglie equation:

$$\lambda = \frac{h}{p} \qquad (1)$$

Where $h = 6.625 \cdot 10^{-34}$ Js, Planck's constant. The momentum can be calculated from the velocity v that the electrons acquire under acceleration voltage U_A:

$$\frac{1}{2} mv^2 = \frac{p^2}{2m} = e\, U_A \qquad (2)$$

The wavelength is thus:

$$\lambda = \frac{h}{\sqrt{2 m_e e\, U_A}} \qquad (3)$$

Where $e = 1.602 \cdot 10^{-19}$ C (the electron charge), and $m = 9.109 \cdot 10^{-31}$ kg (rest mass of electron). At the voltages U_A used, the relativistic mass can be replaced by the rest mass with an error of only 0.5%.

The electron beam strikes a polycrystalline graphite film deposited on a copper grating and is reflected in accordance with the Bragg condition:

$$n\lambda = 2d \sin\theta \qquad (4)$$

Where d is the spacing between the planes of the carbon atoms and θ is the Bragg angle (angle between electron beam and lattice planes). n is diffraction order equals to (1, 2, …).

In polycrystalline graphite the bond between the individual layers (Fig. 1) is broken so that their orientation is random. The electron beam is therefore spread out in the form of a cone and produces interference rings on the fluorescent screen.

The Bragg angle θ can be calculated from the radius of the interference ring but it should be remembered that the angle of deviation α (Fig. 2) is twice as great (i.e, $\alpha = 2\theta$):

$$\sin 2\alpha = \frac{r}{R} \qquad (5)$$

Where R = 65 mm, radius of the glass bulb.
Mathematically:

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

For small angles α, $\cos \alpha = 1$ (for example $\cos 10° = 0.985$):

$$\therefore 2 \sin \alpha = \frac{r}{R}$$

And for small angles θ:

$$\sin \alpha = \sin 2\theta \cong 2 \sin \theta$$

With these approximations we obtain:

$$r = \frac{2R}{d}\, n\, \lambda \qquad (6)$$

The two inner interference rings occur through reflection from the lattice planes of spacing d_1 and d_2 (Fig. 3), for n = 1 in (6).

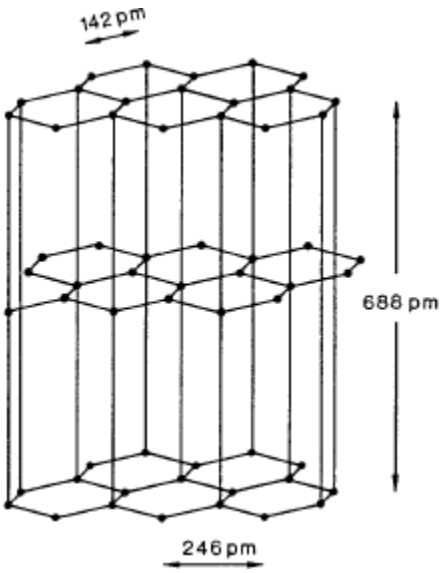


Fig 1 Crystal lattice of graphite

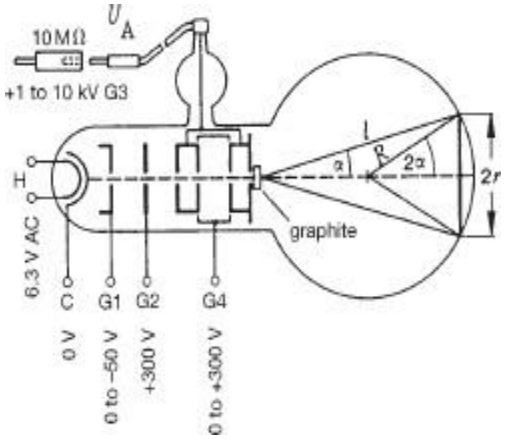


Fig 2: Set-up and geometry of the electron diffraction tube

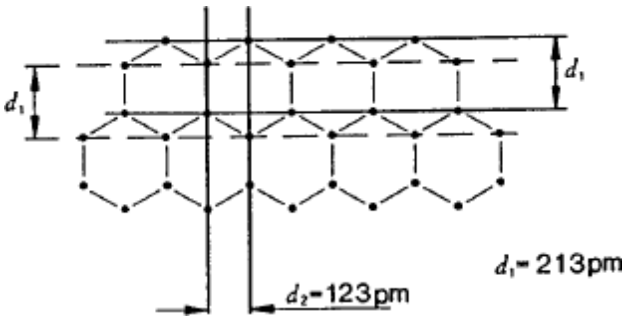


Fig 3 Graphite planes for the first two interference rings

Set-up

Set up the experiment as shown in Fig. 4. Connect the sockets of the electron diffraction tube to the power supply as shown in Fig. 2. Connect the high voltage to the anode G3 through a 10 MΩ protective resistor.

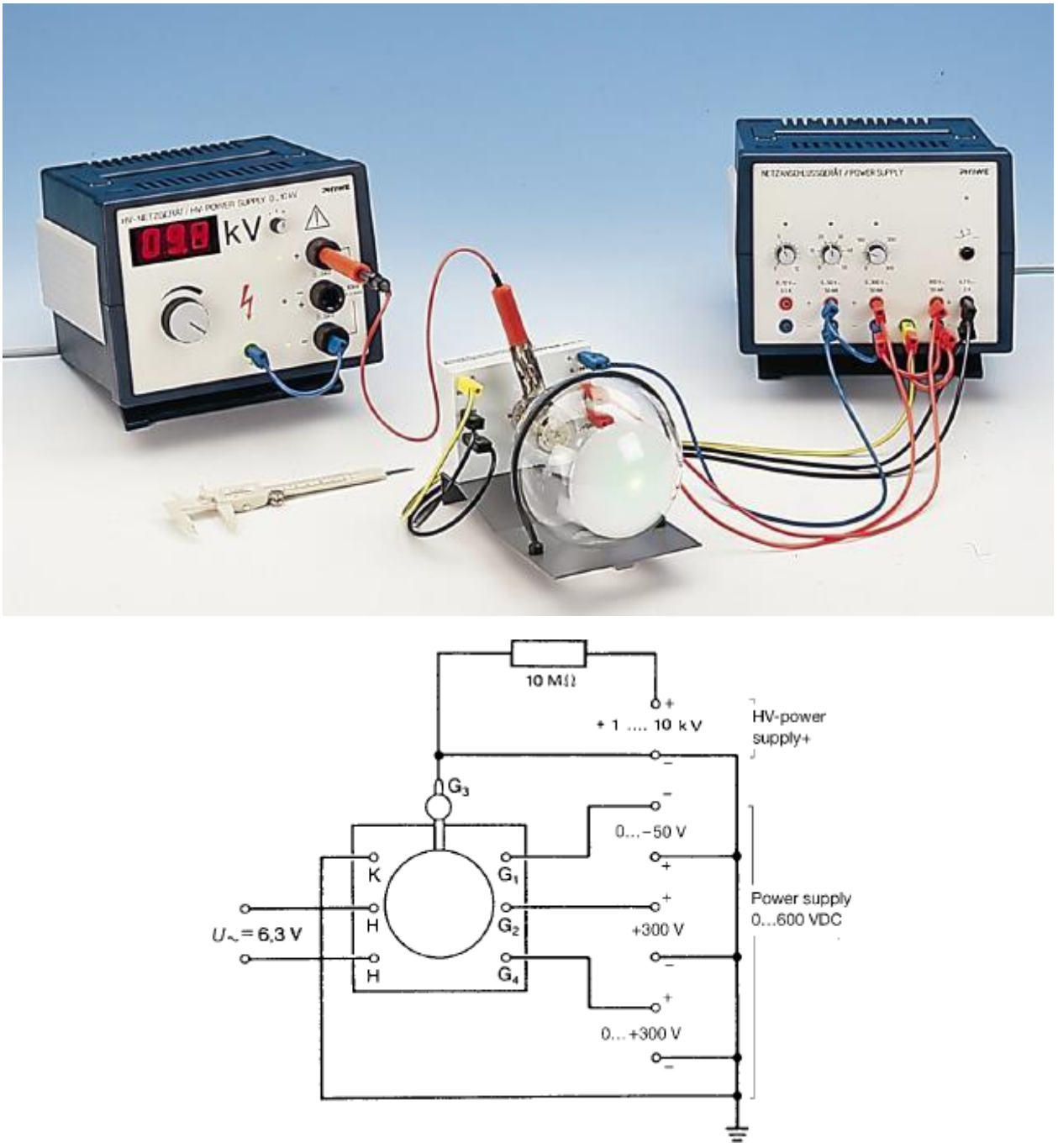


Fig 4 Experimental set-up

Set the Wehnelt voltage G1 and the voltages at grid 4 (G4) and G3 so that sharp, well defined diffraction rings appear.

Procedure

1) Set the anode voltage to 4 kV

2) Measure radius of both the inner and the outer diffraction rings.

3) Increase the anode voltage in steps of 0.5 kV up to 6.5 kV, and repeat step 2 for each step.

4) Tabulate your data as follows:

U_A (kv)	r_1 (mm) (inner ring)	r_2 (mm) (outer ring)	$\lambda\,(pm) = \frac{h}{\sqrt{2m_e e\,U_A}}$
4			
4.5			
5			
5.5			
6			
6.5			

Calculation

1) Plot a graph between the values of rings radii on the y-axis versus the corresponding wavelength values on the x-axis (Fig. 5).

2) Find slope of both lines.

3) Use Eq. 6 for (n = 1) together with each slope to find the corresponding interplanar spacing of the graphite crystal:

$$d_1 = \frac{2R}{Slope_{inner}}$$

$$d_2 = \frac{2R}{Slope_{outer}}$$

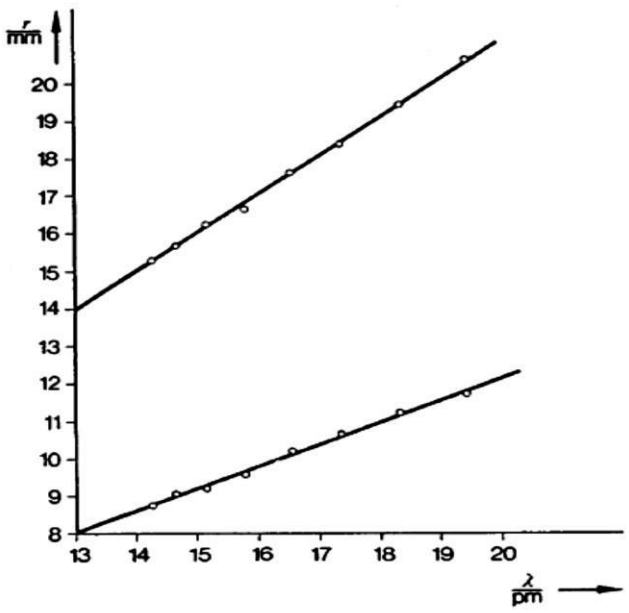


Fig. 5: Radii of the first two interference rings as a function of the wavelength of the electrons.

Questions

1. Compare between electron diffraction and X-ray diffraction.
2. Why radii of the rings decrease with increase of the anode voltage?
3. Why the diffraction pattern appears as a concentric circles?
4. Could neutron diffraction be used to determine the crystal structure?

Experiment (6)

Semiconductor Thermoelectric Generator

Purposes

1- To study the thermoelectric effect in semiconductors.

2- To determine the Seebeck coefficient and efficiency of the thermogenerator.

Theory

Semiconductors are a class of materials that come between the good conductor and the insulator; Germanium and Silicon are examples of semiconductors. These materials were neglected as useful materials for a long time, until transistors were invented in 1948. Since then, these materials have become of great technological importance. Due to their large atomic weight, Silicon and Germanium have much lower resistivity when compared to diamond, as they are made of covalent bonds.

If the bond is sufficiently, weak the thermal vibration of the atoms may occasionally be sufficient to sever a bond even when the temperature is not very high. Breaking a bond does not affect the atoms which it joins together still has three bonds to keep it in place. However, breaking a bond does entail releasing one of the electrons that form it. Thus, at any temperature a small proportion of the large number of bonding electrons are set free. Being no longer confined in the bonds, these electrons may wander freely through the lattice and conduct electric current and heat.

The number of free electrons is essentially fixed at a given temperature, although the process of bond rupture by thermal agitation is continuous. The rate at which electrons are freed is exactly balanced by the rate at which those already free are trapped by an empty bond and lose their freedom. The freed electrons leave behind them gaps in the bonds called holes. These holes move about in much the same random way and at the same velocity as electrons, but with opposite signs.

If a temperature drop is created along a current-free branch of a conductor made up of different semiconductor materials, heat flows from the warmer region to the cooler one. The charge carriers, which take part in this transfer of heat, are unevenly distributed along the conductor. An internal field strength is set up which can be shown to be the e.m.f U_o at the open ends of the conductor (Seebeck effect).

The voltage level depends on the temperature difference of the materials used. To a first approximation, the voltage may be written as:

$$U_o = \alpha(T_h - T_c) = \alpha\,\Delta T \qquad (1)$$

where α is called the Seebeck coefficient of the combination of materials used. T_h is the temperature of the hot side and T_c the temperature of the cold side.

Applying the expression:

$$U_o = a + b\Delta T \qquad (2)$$

From Fig. 1, we get:

$$b = \text{Slope} \qquad (3)$$

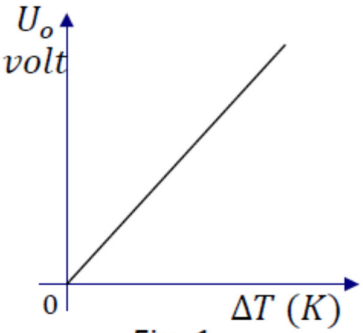


Fig. 1

The thermogenerator consists of 142 elements connected in series. The Seebeck coefficient of the semiconductor combination used is therefore:

$$\alpha = \frac{\text{Slope}}{142} \qquad (4)$$

At any temperature difference ΔT, we can calculate the quantity of heat (Q) flowing through the generator in unit time (t) using the expression:

$$\Delta T = a + bt \qquad (5)$$

$b = \frac{d\Delta T}{dt} = \text{Slope}$ of the graph of temperature difference as a function of time (Fig. 4) and knowing that:

$$\frac{dQ}{dt} = P_{th} = w\frac{d\Delta T}{dt} \qquad (6)$$

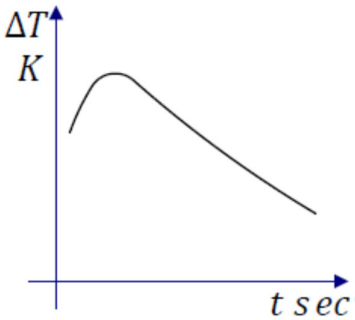


Fig. 4

Where P_{th} is the thermal power, and w is the water equivalent.

If m is the mass of water and c is its specific heat, then:

$$w = m \cdot c \qquad (7)$$

$$P_{th} = w * \text{Slope} \qquad (8)$$

Fig. 5 shows the electrical power generated as a function of the temperature difference. If the thermal power (P_{th}) and Electrical power (P_{el}) is calculated at the same temperature, the efficiency (η) of the generator can be determined from the equation:

$$\eta = \frac{P_{el}}{P_{th}}\% \qquad (9)$$

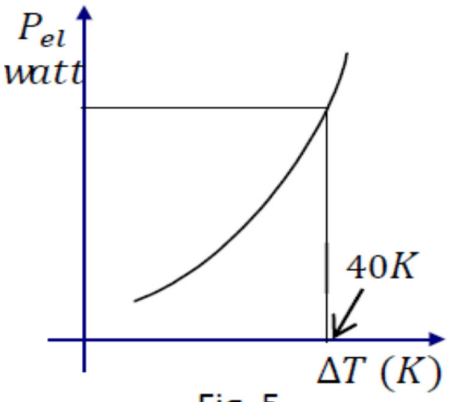


Fig. 5

The thermogenerator (based on the Seebeck effect) produces an electric voltage proportional to the temperature difference between hot and cold junctions. The heat that enters or leaves a junction of a thermoelectric device has two sources (1) the presence of a temperature gradient at the junction (2) the absorption or liberation of energy due to the Peltier effect.

A thermoelectric module generator basically consists of a thermocouple comprising a p-type and n-type semiconductor connected electrically in series and thermally in parallel. The electrical connection allows adding the voltage obtained at each thermocouple due to the Seebeck effect. Therefore, the output voltage of the thermoelectric generator is proportional to the number of thermocouples and the temperature gradient between the cold and the hot side.

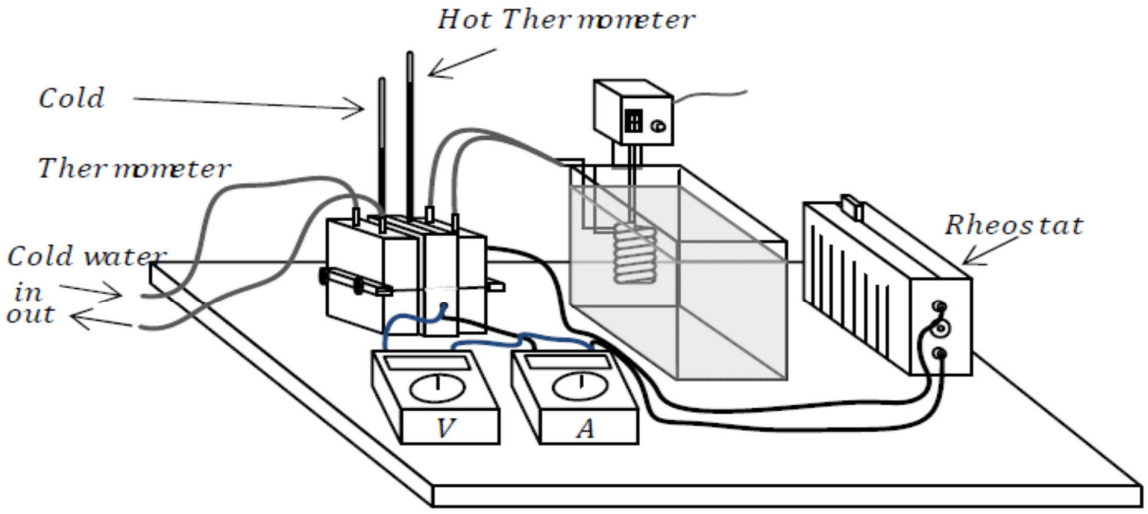


Fig. 6

Procedure

1. Arrange the apparatus as shown in Fig. 6.
2. Fill the cold side with water and let the water flow at a constant rate throughout the experiment.
3. Put the thermometers in their proper positions using the holes in the thermoelectric generator provided for this purpose.
4. Turn on the thermostat heater and wait until the temperature difference between the hot and cold sides reaches 45 C°.
5. Then switch off the thermostat and measure (up to the temperature difference of 35 C°) the temperatures of the hot T_h and the cold T_c sides, current I, and voltage U across an external resistance approximately the same value as the internal resistance as a function of time.
6. Tabulate your measurements such as the following table:

The thermostat is switched off and the system cools down The rheostat fixed at the value of the internal resistance of the thermogenerator						
t (min)	T_h (℃)	T_c (℃)	I (A)	U (v)	$\Delta T = T_h - T_c$	$P_{electric} = I \times U$
0					45	
1						
2						
.						
.					35	

Calculations:

1- Find the Seebeck coefficient (α) of the semiconductor from Fig. 1 and Eq. (4).

2- From the graph shown in Fig. 4 determine the slope of the descending curve at one particular constant temperature difference (for example 40 ℃) by drawing a tangent to the curve or by linear regression, and from Eq. (8) calculate the thermal power as follows:

$$P_{th} = w \cdot Slope$$

The mass (m) of the water in the cold side chamber is 0.194 kg and the specific heat capacity (c) of water is 4182 $\frac{\text{J}}{\text{kg.K}}$. The water equivalent (w), thus, is given by:

$$w = m * c = 0.194 \times 4182 = 811 \text{ J/K}$$

3- From Fig. 5, find the electrical power generator at the same temperature difference at which P_{th} from Fig. 4 was determined (e.g. 40 ℃).

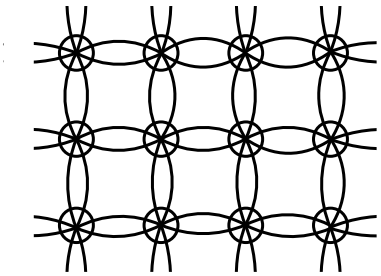
4- Find the efficiency of the thermogenerator using Eq. (9):

$$\eta = \frac{P_{el}}{P_{th}} \%$$

The actual value of η is 0.009 or 0.9 %.

Note

1. The internal resistance of the measuring equipment is disregarded.
2. The tap water must continuously flow at a constant rate throughout the experiment.
3. Although the electrons are moving through the network of bonds, they are not strictly to be regarded as traveling waves. Travelling waves have a constant intensity at every point while we see that in covalent bonds the electron clouds concentrate midway between the atoms. The electron density is more like the distribution in a set of standing waves with nodes at the ion centres and antinodes between them as shown in Fig. 7.



I (mA)

4. The bond strength decreases as the atomic weight increases.
5. The Seiko Thermionic watch is a significant milestone. It uses a thermoelectric generator to convert heat from the wrist into electrical energy. It is the first watch to be powered by a temperature gradient between the body and the environment temperature. The thermoelectric generator produces a power of at least 1 μWatt when the temperature difference is in the range of 1 ℃ − 3 ℃.

Questions

1. Why the sign of the Seebeck coefficient in the $p-$ type semiconductors is positive? What is the sign of the Seebeck coefficient in the $n-$ type semiconductors?
2. What is the difference between the Seebeck effect in metals and semiconductors?
3. Is the Seebeck effect a reversible process?
4. Can we use one type of semiconductor, that is p-type or n-type only, in this experiment?
5. What is the role of the used rheostat in this experiment?
6. What will happen if we use an intrinsic semiconductor?
7. What are the applications of the Seebeck effect?
8. Why are semiconductor materials preferable to metals in the thermoelectric refrigerator?

Appendix (Physical Constants)

Quantity	Value
Elementary charge	$e = 1.60210 \times 10^{-19}$ c
Speed of light in vacuum	$c = 2.997925 \times 10^{8}$ m. sec^{-1}
Planck's constant	$h = 6.6256 \times 10^{-34}$ J.sec
Reduced Planck's constant($\hbar = \boldsymbol{h/2\pi}$)	$\hbar = 1.05450 \times 10^{-34}$ J.sec
Electron rest mass	$m_e = 9.1091 \times 10^{-31}$ kg
Neutron rest mass	$m_n = 1.675 \times 10^{-27}$ kg
Proton rest mass	$m_P = 1.67252 \times 10^{-27}$ kg
Avogadro's number	$N_A = 6.02252 \times 10^{22}$ $mole^{-1}$
Boltzmann's constant	$k_B = 1.38054 \times 10^{-23}$ J. K^{-1}
Universal gas constant	$R = 8.3143$ j. K^{-1}. $mole^{-1}$
Stefan-Boltzmann constant	$\sigma = 5.6697 \times 10^{-3}$ ergs $cm^{-2}sec^{-1}$ deg^{-4}
Coulomb constant	$C_o = 8.988 \times 10^{9}$ Nm^2c^{-2}
Gravitational constant	$G = 6.673 \times 10^{-11}$ Nm^2kg^{-2}
Solar constant	1340 W m^{-2}
Electron volt	$eV = 1.60210 \times 10^{-19}$ J
Velocity of sound in dry air (**at 0** ℃)	331.45 m cm^{-1}
Density of air at standard condition	1.293 kg m^{-3}
Average density of earth	5.518×10^{3} kg m^{-3}
Standard atmospheric pressure (1atm)	760 mmHg
	1.01325×10^{5}pa
	760 torr
	1.013 bar
Mechanical equivalent of heat	4.186 J. cal^{-1}
Permeability of free space	$\mu_o = 4\pi \times 10^{-7}$Wb $A^{-1}m^{-1}$
Permittivity of free space	$\epsilon_o = 8.854188 \times 10^{-12}$ c^2. N^{-1}. m^{-2}
Atomic mass unit	$1u = 1.66054 \times 10^{-27}$ kg
Acceleration due to gravity	g = 9.80665 m. s^{-2}
Rydberg constant	$R_\infty = 1.097 \times 10^{7}$ m^{-1}
Lorenz number	$L = 2.4 \cdot 10^{-8}$ $W\Omega/K^2$